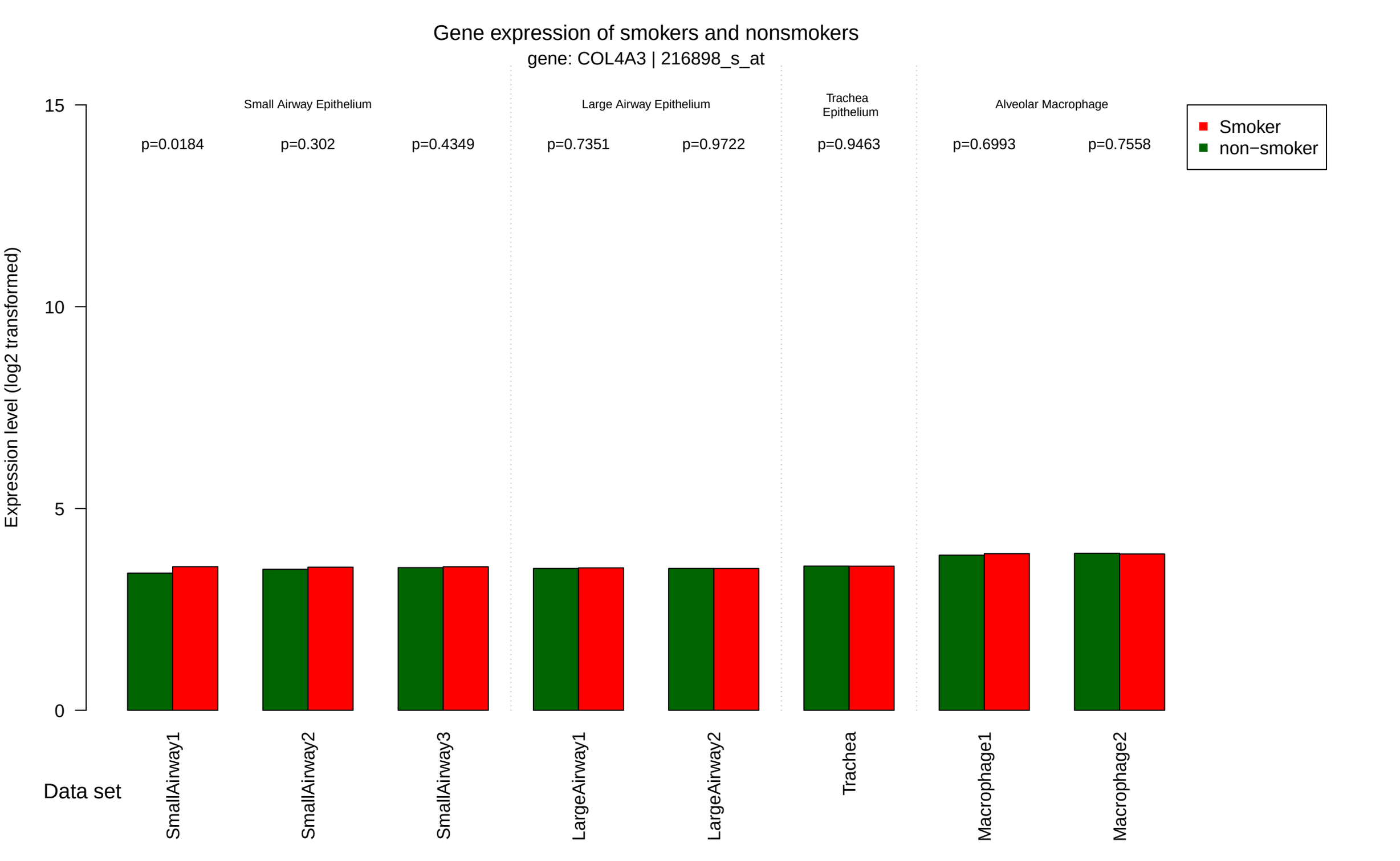
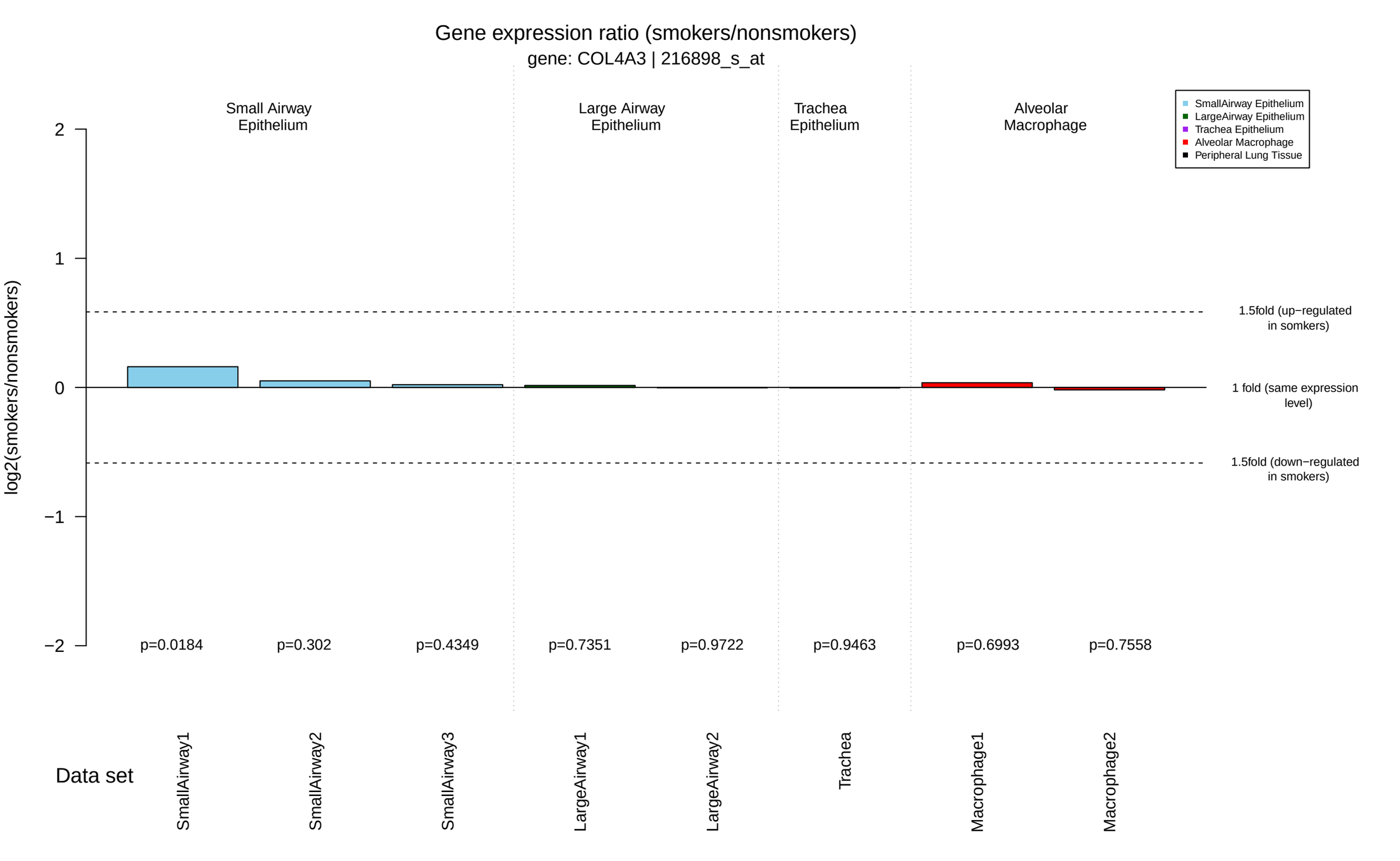
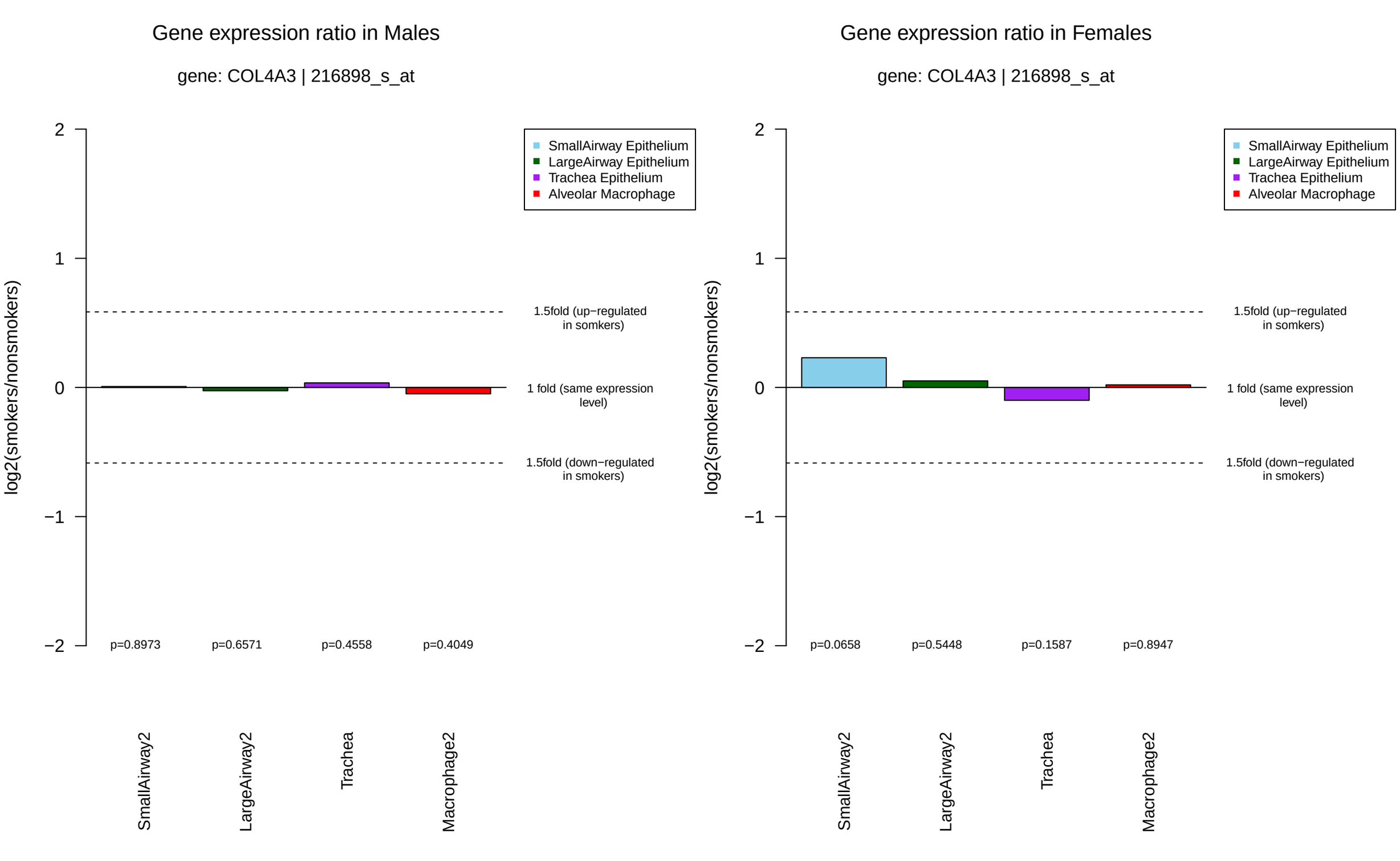
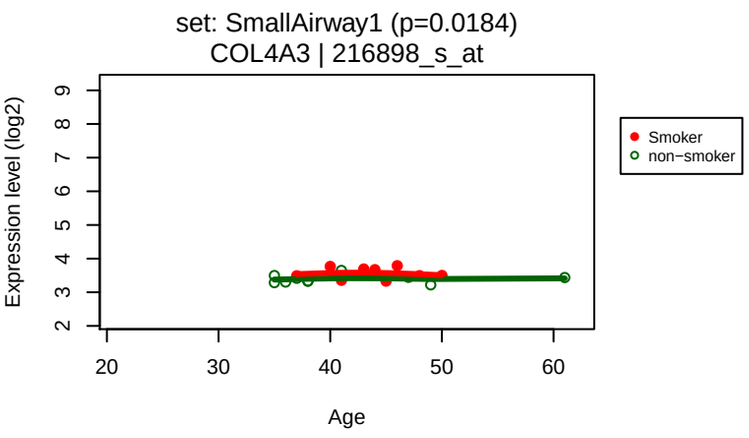
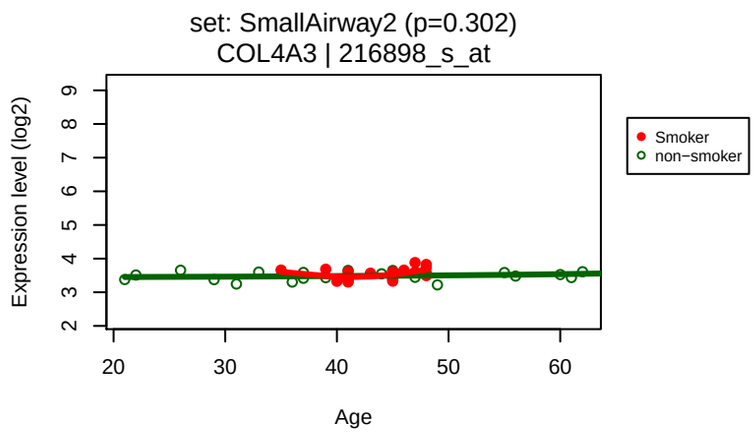
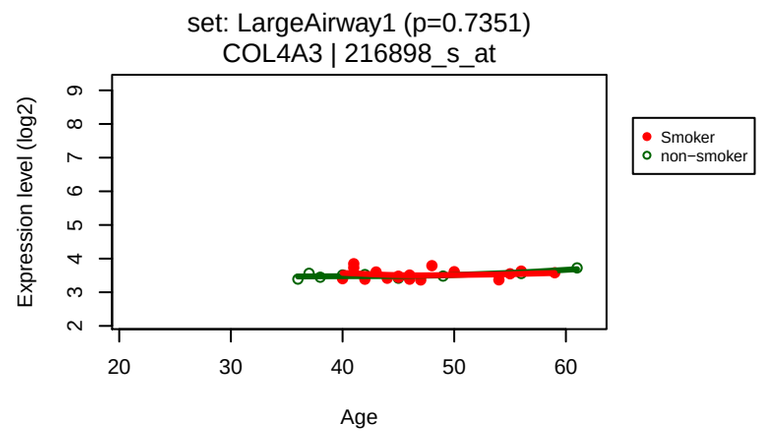

Gene expression of smokers and nonsmokers
gene: COL4A3 | 216898_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0184
p=0.302
p=0.4349
p=0.7351
p=0.9722
p=0.9463
p=0.6993
p=0.7558
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

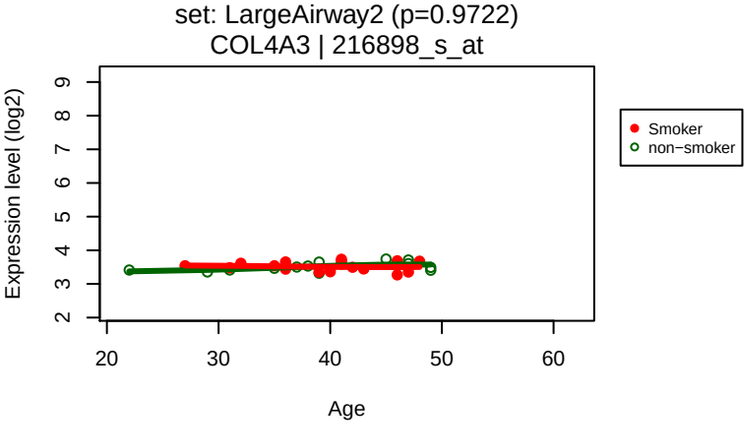

Gene expression ratio (smokers/nonsmokers)
gene: COL4A3 | 216898_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0184
p=0.302
p=0.4349
p=0.7351
p=0.9722
p=0.9463
p=0.6993
p=0.7558
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

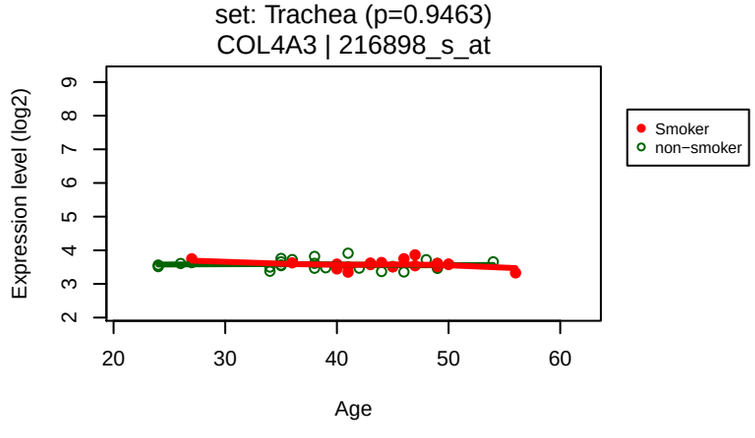

Gene expression ratio in Males
gene: COL4A3 | 216898_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8973
p=0.6571
p=0.4558
p=0.4049
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A3 | 216898_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0658
p=0.5448
p=0.1587
p=0.8947
SmallAirway2
LargeAirway2
Trachea
Macrophage2

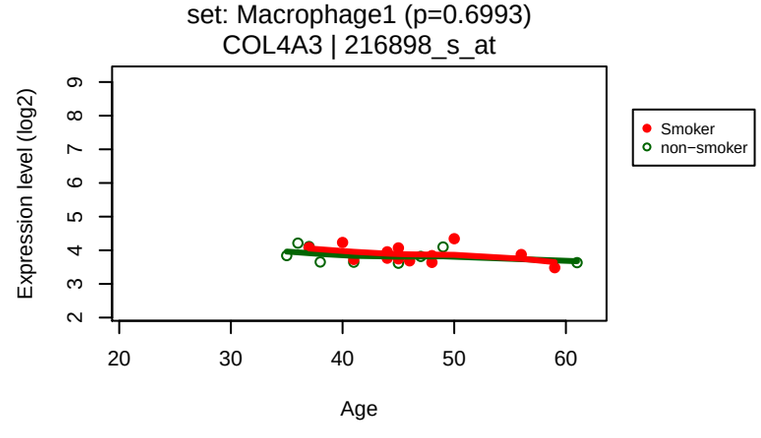

set: SmallAirway1 (p=0.0184)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker

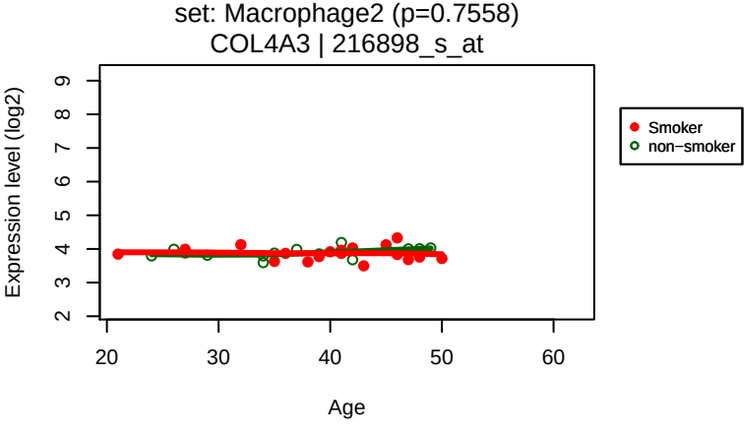

set: SmallAirway2 (p=0.302)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker

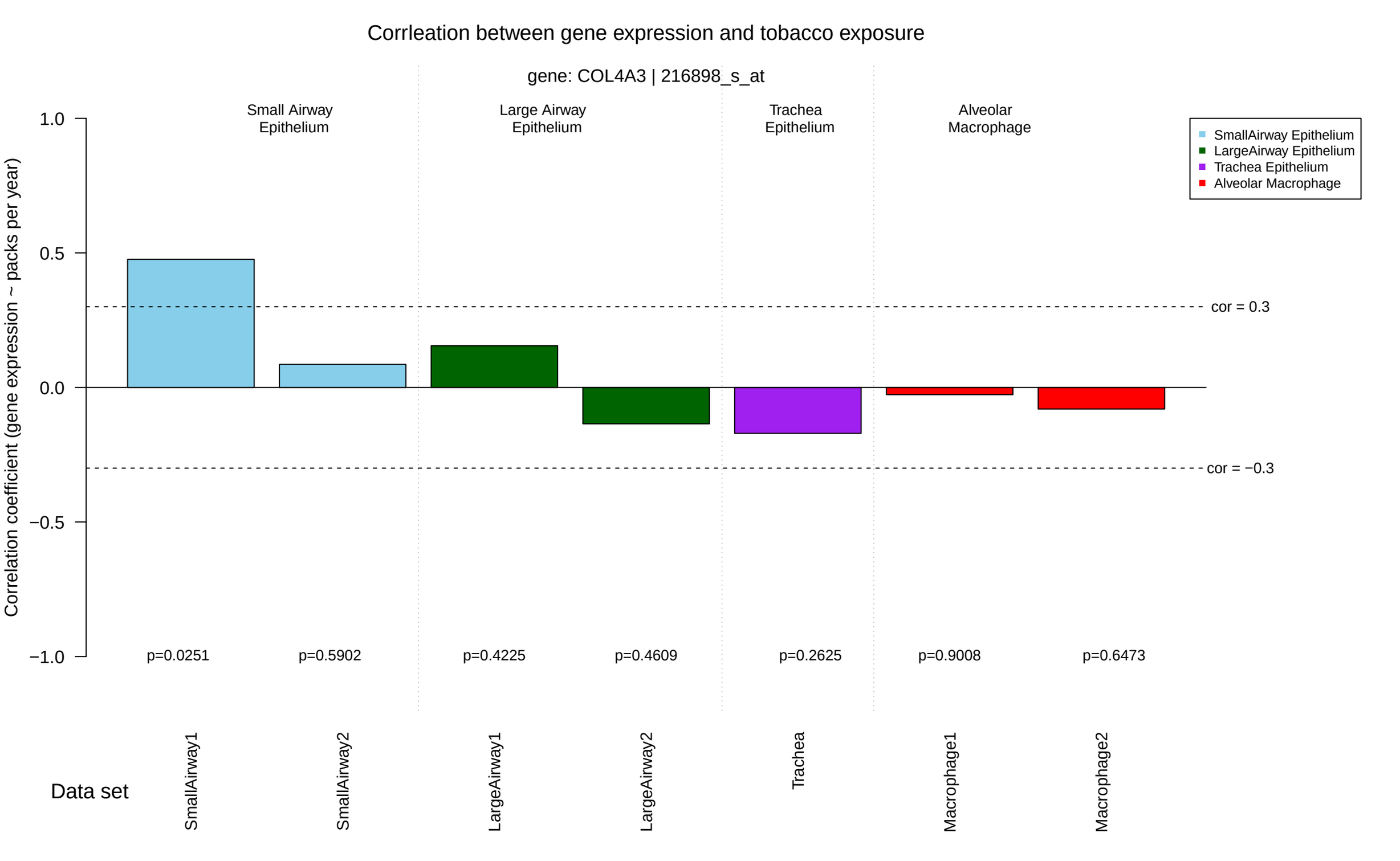

set: LargeAirway1 (p=0.7351)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9722)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9463)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6993)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7558)
COL4A3 | 216898_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A3 | 216898_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0251
p=0.5902
p=0.4225
p=0.4609
p=0.2625
p=0.9008
p=0.6473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2